Andrew Chen

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Technical Skills

- **Languages:** C, Python, Java, HTML, Javascript, React.js
- **Software Development:** Embedded Linux, Payment Terminals, Network Management, API Integration,, Diagnostic Tooling
- **Tools & Methodologies:** Git, Docker, Rest API, Cross Compilation, EMV, ISO 8583

Work Experience

SOFTWARE ENGINEER - UIC PAYMENTS INC. — **OCT 2024 - PRESENT**

- Sole developer for Bezel8-SP unattended payment terminal, owning core embedded C software managing full payment workflows, dynamic device configurations and cloud terminal management.
- Optimized QA feature testing time through the development of on-device diagnostic tools, accelerating provisioning across hundreds of devices.
- Reduced SSL connection handshake time from ~10 seconds to 2–3 seconds by upgrading to modern elliptic curve algorithms, cutting boot time by up to 50%.
- Developed bespoke payment applications, including an unattended car wash flow that managed pulse signals for machine operation, timed re-crediting for uninterrupted machine operation, fully configurable settings, and end-to-end payment flows.
- Diagnosed and fixed embedded payment crashes caused by edge case transaction flow errors, memory handling bugs and startup failures; preventing over $5,000 in RMA shipping and repair costs.
- Designed and implemented Voyager fleet payment processing core within the payment engine, routing required EMV L2 and L3 data, handling ISO 8583 messages, and adding a custom transaction flow to capture driver identification data in compliance with strict official specification requirements.
- Created Interac Canada debit transaction handlers for purchase and refund flows, implementing specification-mandated data fields and debit-specific state handling within the existing payment engine.
- Led entire payment certification efforts for Voyager fleet card and Interac debit, coordinating with certification advisors, executing test cases, resolving kernel issues, and preparing required documentation, achieving full production certification for both payment cores.
- Built network management and configuration flows for Linux-based terminals, supporting DHCP/static IP, DNS, gateway, and subnet settings across multiple hardware generations.

Projects

EYEMATIC - AUTOMATIC IRIS DETECTION — **AUG 2023 - JUNE 2024**

- Customized PolarFire SoC FPGA firmware to support real-time neural network detection of iris, pupil, and surgical instruments with >95% confidence for cataract surgery assistance.
- Trained Tiny-YOLOv3 on 1,000+ annotated frames of surgical footage and generated custom FPGA firmware to support the full inference pipeline, including camera input and HDMI passthrough.

Education

UNIVERSITY OF CALIFORNIA, SANTA BARBARA — **SEP 2020 - JUNE 2024**

- Bachelors of Science in Computer Engineering